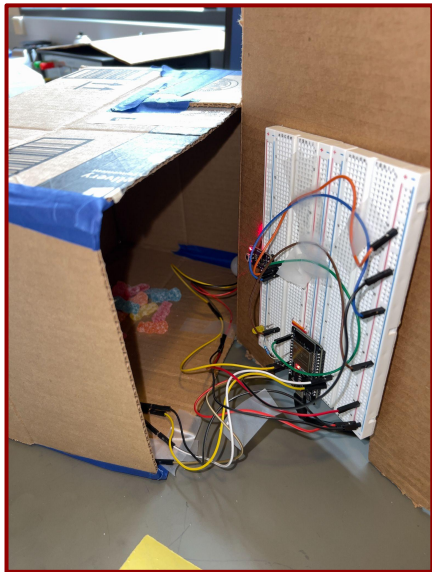


A man in a black t-shirt and gloves is working at a computer in a server room. He is wearing a headset and is looking at the monitor. The room has white walls and a ceiling with square light fixtures. A large orange text overlay is in the center of the image.

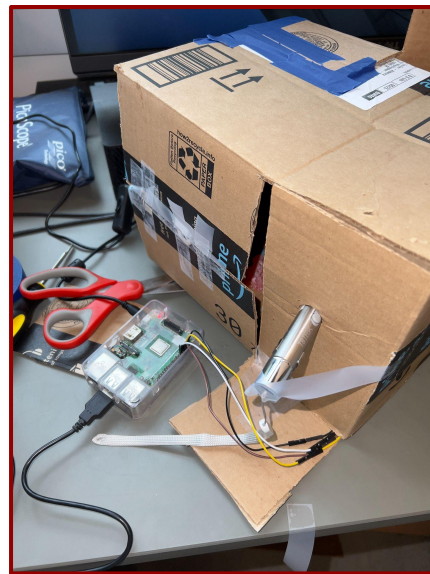
Anti-Mission Impossible Security Room

By Vladimir Andral and Nikunj Shah

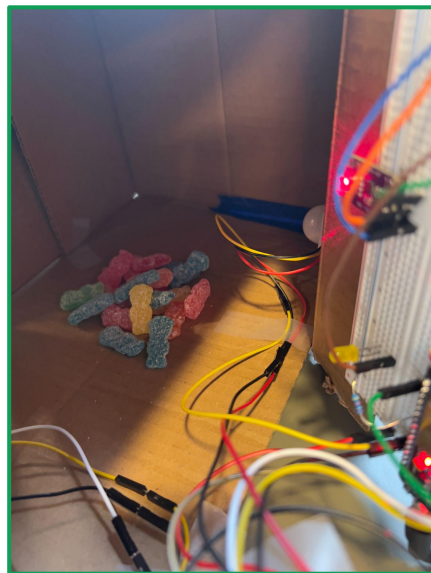
Looking
inside thru
front



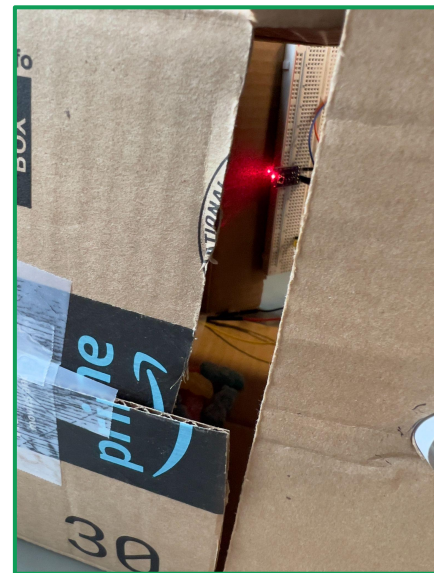
Looking
outside back



Inside

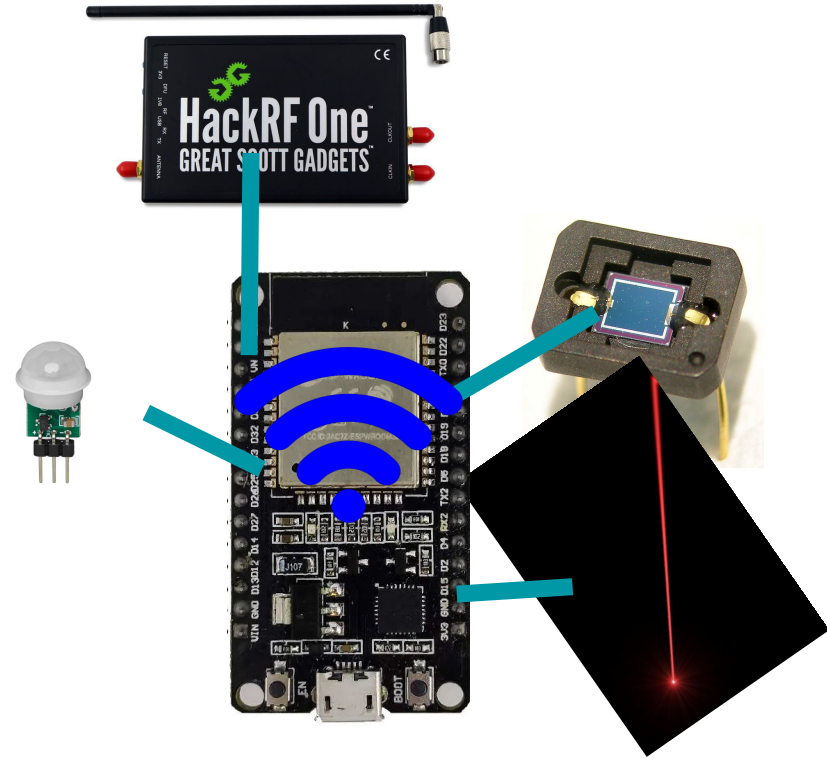


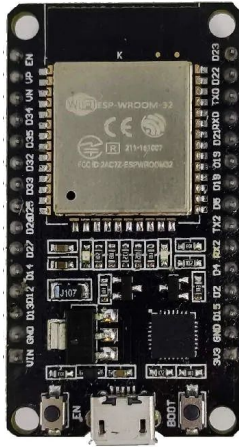
Looking
inside thru
back



Required topics in your presentation

- System components: Introduce all the components/materials used in your system. You **should also include the estimated cost of your system (please also provide a breakdown table).**
- System behavior: Describe and demonstrate the system behavior.
 - Please clearly describe the basic and added system activities, particularly the command and control.
 - Please clearly describe the behaviors of central and peripheral devices.
- Sensors and actuators
 - Please clearly describe what sensors and actuators are used in your system and how they are used.
- Wireless communication
 - Please clearly describe how wireless communication is established and maintained in your system.





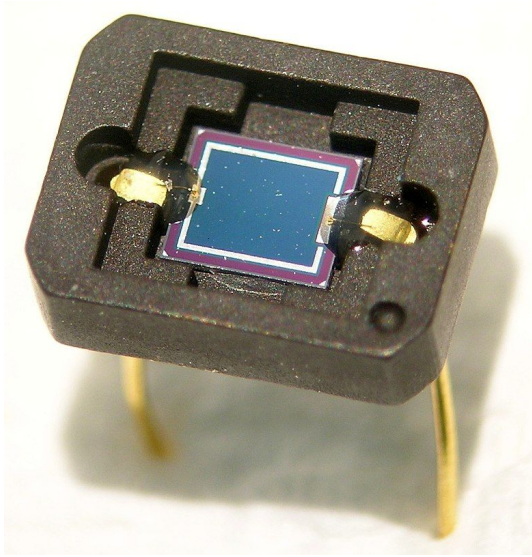
ESP32: Primary and
Secondary
controllers/processors



HackRF One:
Scans radio
frequency band



PIR Sensor:
Detects living
presence (or fire)



Photodiode:
“Wireless” door
movement
detection

System components

Project Component List			
Component	Price	Quantity	Purpose
ESP32 DOIT DevKit V1	\$ 18.99	2	Central and peripheral μ Controllers
HackRF One	\$339.95	1	Senses network traffic on a specific* mobile frequency band
Raspberry Pi Model 3 B+	\$53.59	1	Host device interpreting HackRF data; sends to peripheral ESP32
COINMARK Photodiode	\$6.89	1	Measures change in laser strike
Parallax PIR Sensor (Rev. B)	\$16.95	1	Monitors presence of a human/living being
TBTEEK Laser	\$7.98	1	Shines on photodiode
MAX 9814 Microphone amplifier circuit	\$9.99	1	Amplifies signal coming from photodiode
Total Price: \$444.35 (Spent \$86.80*) *Bought a RasPi Zero W that we did not use			

Sensors and actuators

This system is a type of home security system.

Sensors: We have the PIR sensor(Parallax), a photodiode(COINMARK) that is connected to an amplifier(MAX 9814), and an rf module(HackRF One).

In this system, we use the PIR sensor to see if someone walks inside the “door”. The amplifier is connected to a photodiode to amplify the signal coming from the photodiode. We have a laser pointed at the photodiode and that photodiode detects the change in light. This acts as a trigger when someone opens the “door”. The RF module is connected to a raspberry pi and it just scans the network band to see if there is a signal. (someone connected to the network)

We do not have any actuators.

Commands

```
/*  
| | | target          action          attributeName          attributeValue  
| | | [<sensor>/system]      [set/get/schedule]      [pollRate/sensitivity/Mode]      [<Mode>/<uint>] <uint> <uint>  
| | | [<sensor>/system]      demand  
| | | System behavior is autonomous  
*/
```

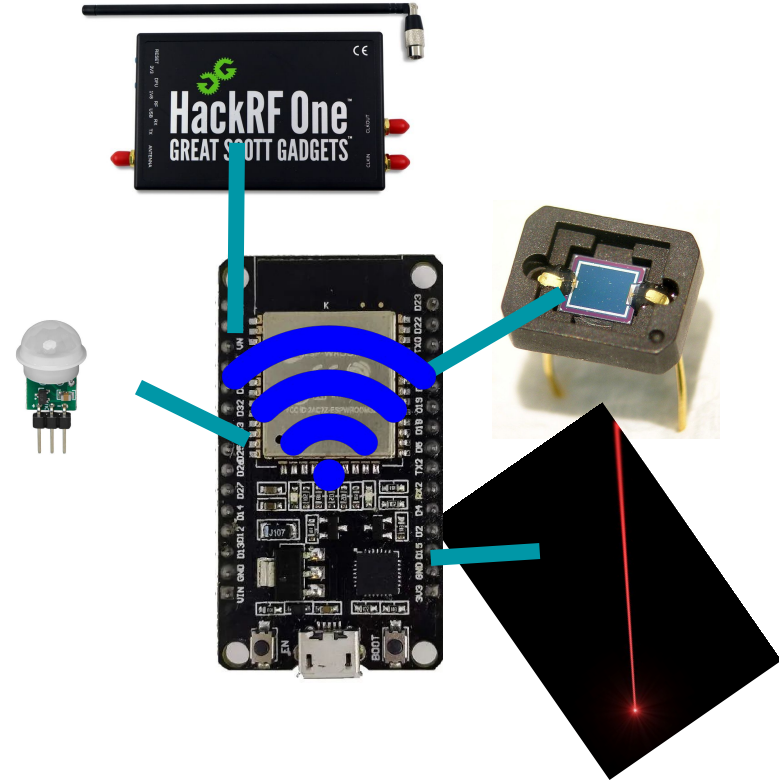
```
String targetNames[] = {  
    "pir",  
    "photo",  
    "rf",  
    "system"  
};
```

```
String actionNames[] = {  
    "demand",  
    "set",  
    "get",  
    "schedule"  
};
```

```
String AttributeNames[]{  
    "pollrate",  
    "sensitivity",  
    "mode"  
};
```

```
String AttributeValues[]{  
    "trig",  
    "poll",  
    "trigpoll",  
    "enable",  
    "disable",  
    "normal",  
    "maintenance",  
    "quiet",  
    "lockdown",  
    "custom"  
};
```

Wireless communication: ESP-NOW



Wireless communication

```
struct __attribute__((packed)) Peri_Msg
{
    bool recv_msg_error = true;

    // Works b/c _INVALID is 0
    ReadingType sensorReadingType[NUM_OF_SENSORS + INVALID_OFFSET] = {READING_INVALID};

    // Works b/c init to 0 and 0 is false
    bool sensorEnabled[NUM_OF_SENSORS + INVALID_OFFSET] = {0};
    // TODO: This does not initialize every element to -1. Must loop through and initialize
    int sensorData[NUM_OF_SENSORS + INVALID_OFFSET] = {-1};
    bool sensorDetected[NUM_OF_SENSORS + INVALID_OFFSET] = {0};
    int numOfDetectInPeriod[NUM_OF_SENSORS + INVALID_OFFSET] = {-1};

    int getResult[NUM_OF_TARGETS + INVALID_OFFSET] = {-1};

    int time[NUM_OF_TIME_COMPONENTS] = {-1};
};
```

Some Of Many Possible System Commands

System set mode disable

System set mode enable

System set mode poll

System set mode trigpoll

System set mode trig

System set mode disable

Pir set pollrate 10

Pir get pollrate

Pir set sensitivity 100

Pir get sensitivity - not possible

Pir set mode normal

Pir get mode

rf set pollrate 10

rf get pollrate

rf set sensitivity 100

rf get sensitivity

rf set mode lockdown

rf get mode - should be poll and true

Pir get mode - disabled

System get mode

System set mode lockdown

System get mode

System set sensitivity 100

System get mode

photo set pollrate 10

photo get pollrate

photo set sensitivity 100

photo get sensitivity

photo set mode quiet

photo get mode - should be trigger and enabled

System set mode normal

System get mode

System set mode quiet

System get mode

System set mode maintenance

System Behavior Combos

System Behavior changed from sensor data

1. If the PIR sensor is in "normal" mode, and it detects movement, then it doubles the pollrate of the RF sensor.
2. If Photodiode detects a change in light (i.e. laser stops pointing at it) then it forces the system to go into lockdown mode.
3. If RF detects network traffic, then increase the sensitivity of the other 2 sensors

System behavior, other than the basic system activities

4. User can schedule changes sensor or system attributes (e.g., system schedule pollrate 2000 12 30)
5. 2 mins of no motion causes the system to cut down the sensor polling rate (Power saving mode) (Interrupts will still be enabled)
6. When the clock hits X time, the system wipes the memory and recalibrates the sensors
7. If the laser hits the photodiode and no motion is detected for 10 seconds, when in LOCKDOWN mode, the system assumes the vault has been successfully secured and goes to NORMAL mode
8. If 2 out of the 3 devices get triggered then turn on yellow led

A note on network traffic sensor

Band	Duplex mode ^[A 1]	f (MHz)	Common name ^[9]	Subset of band	Uplink ^[A 2] (MHz)	Downlink ^[A 3] (MHz)	Duplex spacing (MHz)	Channel bandwidths ^[A 4] (MHz)	Notes
n1	FDD	2100	IMT	n65	1920 – 1980	2110 – 2170	190	5, 10, 15, 20, 25, 30, 40, 45, 50	
n2	FDD	1900	PCS	n25	1850 – 1910	1930 – 1990	80	5, 10, 15, 20, 25, 30, 35, 40	
n3	FDD	1800	DCS		1710 – 1785	1805 – 1880	95	5, 10, 15, 20, 25, 30, 35, 40, 45, 50	
n5	FDD	850	CLR	n26	824 – 849	869 – 894	45	3, 5, 7, 10, 15, 20, 25 ^[B 1]	
n7	FDD	2600	IMT-E		2500 – 2570	2620 – 2690	120	5, 10, 15, 20, 25, 30, 35, 40, 50	
n8	FDD	900	Extended GSM		880 – 915	925 – 960	45	5, 10, 15, 20, 25 ^[B 1] , 35 ^[B 1]	
n12	FDD	700	Lower SMH	n85	699 – 716	729 – 746	30	3, 5, 10, 15	
n13	FDD	700	Upper SMH		777 – 787	746 – 756	–31	5, 10	
n14	FDD	700	Upper SMH		788 – 798	758 – 768	–30	5, 10	
n18	FDD	850	Lower 800	n26	815 – 830	860 – 875	45	5, 10, 15	Japan
n20	FDD	800	Digital Dividend		832 – 862	791 – 821	–41	5, 10, 15, 20	EU
n24	FDD	1600	Upper L-band		1626.5 – 1660.5 ^[B 2]	1525 – 1559 ^[B 3]	–101.5	5, 10	US
n25	FDD	1900	Extended PCS		1850 – 1915	1930 – 1995	80	5, 10, 15, 20, 25, 30, 35, 40, 45 ^[B 1]	
n26	FDD	850	Extended CLR		814 – 849	859 – 894	45	3, 5, 7, 10, 15, 20, 25 ^[B 4] , 30 ^[B 4]	
n28	FDD	700	APT		703 – 748	758 – 803	55	3, 5, 10, 15, 20, 25, 30	

Check Active Cellular Bands (4G/LTE/5G)

- **iPhone:** Open the Phone app, dial *3001#12345* and press Call. Tap the menu, navigate to **Serving Cell Info**, and look for the **Band Info** or **Freq Band Indicator**. www.tp-link.com +1
- **Samsung Galaxy:** Open the Phone app and dial *#0011#. This opens **Service Mode**, where your current LTE or 5G Band will be displayed clearly on the screen. [Stella Doradus](#)

RADIO SERVICES COLOR LEGEND			
	AEROMOBILE		INTER SATELLITE
	AEROMOBILE MOBILE		MOBILE AEROMOBILE
	AEROMOBILE MOBILE SATELLITE		RADIO TERMINATOR SATELLITE
	AEROMOBILE WINDINGSTATION		MOBILE MOBILE
	AMATEUR		RADIO COMMUNICATION SATELLITE
	AMATEUR SATELLITE		RADIO COMMUNICATION SATELLITE
	BROADCASTING		RADIO COMMUNICATION SATELLITE
	BROADCASTING SATELLITE		RADIO COMMUNICATION SATELLITE
	EARTH EXPLORATION SATELLITE		RADIO COMMUNICATION SATELLITE
	FIXED		RADIO COMMUNICATION SATELLITE
	FIXED SATELLITE		RADIO COMMUNICATION SATELLITE

GOVERNMENT EXCLUSIVE

NON-GOVERNMENT EXCLUSIVE

SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letter

 U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
October 2003



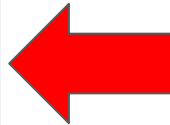
Dashboard

Carrier: Verizon
Network Capabilities: SA+NSA
TAC: 513
PLMN: 311 480
Phone Number: +12065611421

LTE

Band 66
Cell Id 77432844
PCI 414
RSRP -107 dBm
RSRQ -10 dB
SINR0 4.9 dB
SINR1 5.9 dB

Other LTE Bands



Band	Duplex mode ^[A 1]	f (MHz)	Common name ^[9]	Subset of band	Uplink ^[A 2] (MHz)	Downlink ^[A 3] (MHz)	Duplex spacing (MHz)	ba
n66	FDD	1700 2100	Extended AWS		1710 – 1780	2110 – 2200 ^[B 8]	400	5, 10

Questions?
Let's Demo

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```